

Q3LOCK Universal Orbit-Smear Carrier and Ground-Doublet Route Split

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This checkpoint records EXP-000803 and advances reusable result R-167 to v1.6 without allocating a new result number. The stable result identifier remains PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT. The v1.0–v1.5 weighted-energy, graph, topology, canonical fixed-temperature OS-mixture and selected-tangent results remain in force with their registered boundaries.

There are two deliberately separate positive conclusions. First, an exact KMS modular right-context estimate and a recursive right-to-left Fejer smoothing argument upgrade the selected EXP-000781 tangent-net theorem from one cyclic character to every fixed finite raw configuration-orbit word moment and Gram block. The conclusion is pointed finite-core Fell/GNS convergence only.

Second, the complete family of exact zero-source finite periodic Hamiltonians defines a beta- and state-independent universal L^1 orbit-smear C-star carrier with one point-norm continuous time shift. The approximate broken doublets from EXP-000789 have two distinct weak-star cluster states on this carrier, and the negative-Arveson estimate proves that both are ground states for its common shift.

The second construction is categorical and nonlocal. It is not a quasi-local thermodynamic limit: its finite-volume quotient maps are not an inductive family, raw configuration characters need not belong to it, and temporal smears need not be spatially local. No all-exhaustion Cauchy, canonical momentum/full CCR carrier, polynomial local generator, phase-KMS quotient identification, broken-sector GNS gap, physical mass gap, continuum, physical empty-space comparison, C6, CP1, Sector A or Pre-A closure is claimed.

2 Provenance identifiers

- Result: PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT, v1.6, R-167.
- Exploration: EXP-000803.
- Closed subgate: PA-CP1-ST8-Q3LOCK-SELECTED-TANGENT-RAW-FINITE-ORBIT-WORD-MOMENT-COMPLETION.
- Closed subgate: PA-CP1-ST8-Q3LOCK-ZERO-SOURCE-FINITE-HAMILTONIAN-L1-ORBIT-SMEAR-CSTAR-CARRIER.
- Closed subgate: PA-CP1-ST8-Q3LOCK-UNIVERSAL-ORBIT-SMEAR-DISTINCT-ALGEBRAIC-GROUND-DOUBLET.

- Successor: [PA-CP1-ST8-Q3LOCK-QUASI-LOCAL-RAW-OSCILLATOR-ALL-EXHAUSTION-COMMON-ALPHA-AND-BROKEN-GNS-GAP](#).
- New negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-STATIC-TAIL-ONLY-PROJECTED-ORBIT-LOCALITY](#).
- Retained negative: [NG-2026-08-09-PRE-A-ST8-Q3LOCK-POSTHOC-DIRECT-SUM-COMMON-DYNAMICS](#).
- Retained negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-FIXED-BETA-ENVELOPE-AUTOMATIC-CROSS-BETA-GLUING](#).

3 Selected-tangent finite raw-word completion

Fix $\beta > 0$ and the already registered plus/minus source tangent diagonal net of **EXP-000781**. The v1.5 theorem gives convergence of every fixed finite bandlimited word Gram block and, for each rational finite-support configuration label ξ , the cyclic Fejer error

$$\epsilon_\xi(R) = \frac{\|\xi\|}{\sqrt{\beta\chi}} \left(\frac{2}{R^2} + \frac{\beta\hbar}{R} \right)^{1/2}. \quad (3.1)$$

Use the support-reduced faithful KMS standard form, with the convention $\sigma_s = \alpha_{-\beta\hbar s}$. If C is entire analytic and Y is a bounded left factor, Tomita's right-action identity gives

$$YC\Omega = J\sigma_{-i/2}(C^*)JY\Omega. \quad (3.2)$$

Consequently

$$\|YC\Omega\| \leq \|\sigma_{i/2}(C)\| \|Y\Omega\|. \quad (3.3)$$

If the physical Arveson spectrum of C lies in $[-S, S]$, the standard strip estimate yields

$$\|\sigma_{i/2}(C)\| \leq e^{\beta\hbar S/2} \|C\|. \quad (3.4)$$

For a fixed word $A_1 \cdots A_m$, let C_j be a Fejer-smoothed version of A_j with bandwidth R_j , and put $Y_j = A_j - C_j$. Replace factors from right to left. When the already smoothed right context has bandwidth

$$S_j = \sum_{k>j} R_k,$$

the j -th telescoping error is bounded by

$$e^{\beta\hbar S_j/2} \epsilon_{\xi_j}(R_j). \quad (3.5)$$

Choose R_m first. Once it is fixed, choose R_{m-1} large enough for its finite right-context cost; continue recursively through R_1 . Every finite sum of errors in (3.5) can therefore be made arbitrarily small. Combining this with the fixed-band theorem proves convergence of every fixed finite raw configuration-orbit word moment and Gram entry on the selected tangent net. Independent limiting pivots give pointed finite-core Fell/GNS convergence on the algebraic raw orbit-word span, equivalently an identification of its cyclic subspace in a GNS ultraproduct.

The bandwidths may grow very rapidly toward the left. This proof is not uniform in word length and does not give an arbitrary bounded-context multiplier theorem, common-Hilbert operator strong-star convergence or all-exhaustion uniqueness.

4 Universal zero-source orbit-smear carrier

Let $\Xi_{\mathbb{Q}}$ be the finite-support rational configuration labels on $\mathbb{Z}^3 \times \{1, \dots, 8\}$. For every finite periodic volume Λ , periodize ξ and set

$$W_{\Lambda, \xi} = \exp(i\xi \cdot q), \quad \alpha_t^\Lambda(A) = e^{itH_\Lambda(0)/\hbar} A e^{-itH_\Lambda(0)/\hbar}. \quad (4.1)$$

Only the exact zero-source Hamiltonians $H_\Lambda(0)$ enter this construction. For $f \in L^1(\mathbb{R})$ introduce a symbol $A_{\xi, f}$, linear in f , with

$$A_{\xi, f}^* = A_{-\xi, \bar{f}}, \quad \pi_\Lambda(A_{\xi, f}) = \int_{\mathbb{R}} f(t) \alpha_t^\Lambda(W_{\Lambda, \xi}) dt. \quad (4.2)$$

The integral is a weak-operator integral and satisfies the volume-uniform bound

$$\|\pi_\Lambda(A_{\xi, f})\| \leq \|f\|_1. \quad (4.3)$$

Take the supremum of all finite-volume representation norms on the unital formal star algebra, quotient the common kernel, and complete. Denote the result by $\mathcal{A}_H^0$. The product of all π_Λ is faithful, although no individual quotient representation need be faithful.

Define

$$\theta_s A_{\xi, f} = A_{\xi, \tau_s f}, \quad (\tau_s f)(t) = f(t - s). \quad (4.4)$$

Finite-volume equivariance makes θ_s isometric. The L^1 translation theorem gives point-norm continuity, so $(\mathcal{A}_H^0, \theta)$ is a beta- and state-independent C-star dynamical system. On the smooth smear core its generator is exactly

$$\delta_H A_{\xi, f} = -A_{\xi, f'}. \quad (4.5)$$

Every finite-volume Gibbs state pulled back through π_Λ is a β -KMS state of θ , and fixed- β weak-star accumulation points remain KMS. This is a common categorical carrier theorem, not an identification of the finite Hamiltonian quotients with one spatial thermodynamic limit or with the phase envelopes of EXP-000800.

5 Two distinct ground-state clusters

Use the zero-source periodic doublets $\Psi_L^\pm$ of EXP-000789. Write

$$K_L = H_L(0) - E_{0, L} \geq 0, \quad \varepsilon_L = \langle \Psi_L^\pm, K_L \Psi_L^\pm \rangle. \quad (5.1)$$

If $V = L^3$ and $m_L^2 = \langle S_L^2 \rangle / V^2$, then eventually $m_L \geq m_0 = \sqrt{\rho_*/2}$ and

$$\varepsilon_L \leq \frac{\hbar^2}{4\chi V m_L^2} \leq \frac{\hbar^2}{2\chi V \rho_*}. \quad (5.2)$$

At one fixed site let

$$X = \sum_{e=1}^8 q_e = \sqrt{8} Q, \quad M_3 = (64 M_{4, Q})^{3/4}, \quad (5.3)$$

where the local fourth moment is uniformly bounded on the doublets. Choose one fixed rational $r > 0$ such that

$$r^2 M_3 \leq 3\sqrt{8} m_0, \quad \xi = r(1, \dots, 1). \quad (5.4)$$

The rational label is essential; using the normalized diagonal unit vector would introduce irrational entries. From $|\sin z - z| \leq |z|^3/6$ and parity,

$$\omega_L^+(\sin rX) \geq d, \quad \omega_L^-(\sin rX) \leq -d, \quad d = \frac{r\sqrt{8}m_0}{2} > 0 \quad (5.5)$$

for all sufficiently large L .

Fix $T > 0$ and the triangular function

$$f_T(t) = T^{-1}(1 - |t|/T)_+, \quad \int f_T = 1, \quad \int |t|^{1/2} f_T(t) dt = \frac{8\sqrt{T}}{15}. \quad (5.6)$$

For every contraction B , the near-ground energy estimate gives

$$|\omega_L^\pm(\alpha_t^L(B)) - \omega_L^\pm(B)| \leq 2\sqrt{\frac{2|t|\varepsilon_L}{\hbar}}. \quad (5.7)$$

The single self-adjoint abstract element

$$b = \frac{A_{\xi, f_T} - A_{-\xi, f_T}}{2i} \in \mathcal{A}_H^0 \quad (5.8)$$

therefore obeys the smear error bound

$$\frac{16}{15}\sqrt{\frac{2T\varepsilon_L}{\hbar}} \leq \frac{16}{15}\sqrt{\frac{\hbar T}{\chi V \rho_*}} \longrightarrow 0. \quad (5.9)$$

Thus every joint weak-star cluster pair satisfies $\omega^+(b) \geq d/2$ and $\omega^-(b) \leq -d/2$ and is distinct.

For the Fourier convention

$$\hat{f}(\nu) = \int_{\mathbb{R}} e^{+i\nu t} f(t) dt, \quad (5.10)$$

let $a \in \mathcal{A}_H^0$ have θ -Arveson spectrum in $(-\infty, -\nu]$, with $\nu > 0$ in inverse-time units. Finite-volume equivariance gives

$$\pi_L(a)P_{[0, \hbar\nu)}(K_L) = 0, \quad \omega_L^\pm(a^*a) \leq \frac{\|a\|_H^2 \varepsilon_L}{\hbar\nu} \longrightarrow 0. \quad (5.11)$$

Every cluster state is therefore a ground state of $(\mathcal{A}_H^0, \theta)$. This proves algebraic ground states only on the universal orbit-smear carrier; it proves neither a quasi-local oscillator ground representation nor a positive GNS or physical mass gap.

6 Why the carrier does not close common alpha

The exact finite-dimensional pair

$$(\beta_1, H_1) = (1, -\sigma_x), \quad (\beta_2, H_2) = (2, -2\sigma_x) \quad (6.1)$$

with common configuration label σ_z has orbit frequencies 2 and 4. Its L^1 orbit smears generate $M_2 \oplus M_2$ and carry one abstract C-zero shift with both KMS pullbacks, while the labelled Hamiltonian generators differ by more than a scalar. Hence a universal orbit-smear carrier can package inequivalent quotient dynamics. Carrier existence does not imply a quasi-local thermodynamic Hamiltonian limit, exhaustion uniqueness or a canonical raw local generator. This preserves, rather than overturns, the registered posthoc-direct-sum and automatic-cross-beta-gluings warnings.

7 Projected-tail corridor audit

For a degree- z graph with bounded pair interactions $\|\Phi_e^L\| \leq J_L$, a conservative pairwise-union cutoff estimate is

$$8\sqrt{2}|X||A|e^{\nu_L T} \frac{(\nu_L T)^R}{R!}, \quad \nu_L = \frac{4zJ_L}{\hbar}. \quad (7.1)$$

For the coordinate cutoff $J_L \leq 4cL^2$ and $z = 6$, so $\nu_L \leq 96cL^2/\hbar$. Taking $L = R^\alpha$ makes (7.1) vanish on every compact real-time interval for $0 < \alpha < 1/2$. This is a theorem about the bounded cutoff dynamics only.

Static tail estimates do not remove the cutoff. An exact four-dimensional fixture uses

$$r_n = (2n+1)\pi, \quad \epsilon_n = e^{-r_n^4}, \quad Z_n = 1 + 3\epsilon_n, \quad \rho_n = Z_n^{-1} \text{diag}(1, \epsilon_n, \epsilon_n, \epsilon_n), \quad (7.2)$$

and

$$q_x = r_n \text{diag}(0, 0, 1, 1), \quad q_y = r_n \text{diag}(0, 1, 0, 1), \quad W = \text{diag}(1, 1, -1, -1), \quad X = q_x q_y. \quad (7.3)$$

With $k = \pi\hbar/(4T)$, let $K_{03} = -ik$, $K_{30} = ik$ and all other entries zero; put $H = K + X$. Direct calculation gives

$$\|X\|_D^2 = \frac{r_n^4 \epsilon_n}{Z_n} \rightarrow 0, \quad [\log \rho_n, X] = 0, \quad (7.4)$$

but, for $B = \alpha_T^K(W)$,

$$\|[X, B]\|_D^2 = \frac{2(1 - \epsilon_n)}{Z_n} \rightarrow 2, \quad (7.5)$$

and the full-versus-cutoff orbit two-sided distance tends to 2. The coordinate Gaussian moments are bounded for every fixed coefficient. Thus static Gaussian tails and a vanishing first modular derivative do not imply projected real-time orbit locality.

This fixture is not a Q3LOCK counterexample: ρ_n is not asserted to be Gibbs invariant for H or K . It rejects only that proof inference. A connected two-orientation dynamic tail or quasi-invariance estimate is still required.

8 Devil's-advocate review

1. The modular sign in (3.2) was checked in a finite-density standard-form fixture. The right-context cost uses $\sigma_{i/2}(C)$ and the physical bandwidth factor $e^{\beta\hbar S/2}$.
2. The Fejer replacements are ordered from right to left. One common bandwidth does not close the exponentially weighted estimate; the recursive finite-word choice does.
3. The universal norm is the supremum of exact finite-volume representation norms and is finite by (4.3). Its existence is not confused with inductive compatibility or spatial locality.
4. The zero-source restriction is explicit. Including source-dependent Hamiltonians in the same categorical family would merely package different generators and would not prove source-independent dynamics.
5. The ground-state Fourier sign and energy denominator are fixed by (5.10): negative transfer frequency costs the positive energy $\hbar\nu$ in (5.11).

6. The order witness uses the rational vector $r(1, \dots, 1)$, not the irrational normalized diagonal vector. The constants $8/15$ and $16/15$ are independently recomputed.
7. The four-dimensional static-tail fixture rejects one implication only. It is not a Gibbs Q3LOCK non-Cauchy example and does not prove nonexistence of the sought dynamics.
8. Neither pointed finite-core Fell/GNS convergence nor the categorical carrier is promoted to common-Hilbert operator strong-star, all-exhaustion, quasi-local, momentum, GNS-gap or physical conclusions.

9 Route decision

Three scoped subgates are closed: selected-tangent fixed finite raw-word moments, the zero-source universal finite-Hamiltonian orbit-smear carrier, and two distinct algebraic ground states on that carrier. The projected corridor still lacks a dynamic tilted-tail estimate.

The live successor is [PA-CP1-ST8-Q3LOCK-QUASI-LOCAL-RAW-OSCILLATOR-ALL-EXHAUSTION-COMMON-ALPHA-AND-BROKEN-GNS-GAP](#). It must construct a noncollapsing raw or resolvent oscillator carrier with a spatial local net, prove pairwise all-shape exhaustion Cauchy, identify the finite Hamiltonian generator and phase-KMS quotient representations, and only then determine the broken-sector GNS gap. Until those obligations close, the present ground states are ground states of the categorical carrier only.

10 Reproduction

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_universal_orbit_smear_ground_
doublet_route_split.py
```

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_universal_orbit_smear_ground_
doublet_route_split_independent.py
```

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_universal_orbit_smear_ground_
doublet_route_split_verify.py
```

11 Result footer

Result ID:	PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:	R-167 v1.6; no new result number.
Exploration:	EXP-000803.
Precise statement:	Selected tangent-net convergence for every fixed finite raw orbit-word moment and Gram block; one beta-independent zero-source finite-Hamiltonian L1 orbit-smear C-star carrier; two distinct ground states of its common C0 shift.
Scope:	T0 exact fixed-beta selected-net theorem plus an independent categorical nonlocal carrier theorem.
Dependencies:	EXP-000789, EXP-000801--EXP-000803; R-167 v1.0--v1.5; exact zero-source finite Hamiltonians.
Evidence grade:	ANALYTIC + EXACT EXECUTED + INDEPENDENT + PDF QA.
Reproduction command:	Run the three commands under Reproduction in the listed order, with the integrated verifier last.
Expected output:	Primary PASS 128/128; independent PASS 113/113; integrated PASS 137/137.

Falsification gate: Any modular sign, bandwidth order, norm completion, Fourier sign, ground-energy denominator, rational witness, static-tail limit, hash or scope mismatch.

Tier before / after: C6 T1 / C6 T1; no claim or tier change.

No-overclaim statement: Quasi-local raw oscillator algebra, spatial net, all-exhaustion common alpha, phase-KMS quotient, canonical momentum/full CCR, polynomial generator, broken-sector GNS or physical gap, continuum, physical empty-space sign, C6, CP1, Sector A and Pre-A remain open.

Next required action: Build a spatial raw/resolvent carrier with direct all-shape exhaustion Cauchy and Hamiltonian/KMS quotient identification; then test the broken GNS spectral gap.